Quantum-Assisted Multi-Asset Portfolio Construction

Step 01

Financial and Mathematical Formulation

From portfolio economics to a convex, auditable optimization model

Project technical report

Contents

1 The story: from a list of assets to an investable decision

A portfolio optimizer is often introduced as a simple trade-off: choose assets with high expected returns and low covariance. That description is mathematically useful, but it is not enough for an implementable portfolio. A real decision must also answer:

- How much can be invested in any one asset or asset class?
- How much trading is allowed relative to the current portfolio?
- Can the proposed trades be executed without using an unrealistic share of market liquidity?
- What happens in equity, rate, credit, commodity, and deleveraging stress scenarios?
- How are growth, income, risk control, and trading cost combined without mixing incompatible units?

The project therefore begins with an *exact continuous portfolio model*. The quantum component is not introduced until a feasible classical portfolio exists. This separation is important: the continuous optimizer owns the economic truth, while the later binary model only helps choose a small active trading sleeve.

2 Notation and economic meaning

There are $N = 50$ assets. The main variables and inputs are:

Symbol	Meaning
$w \in \mathbb{R}^N$	Final portfolio weights; $w_i = 0.03$ means 3% of capital in asset i.
$w^0 \in \mathbb{R}^N$	Current or incumbent portfolio weights.
$p, n \in \mathbb{R}_+^N$	Purchase and sale weights.
$g, y \in \mathbb{R}^N$	Expected capital-growth return and expected income yield.
$\Sigma \succeq 0$	Annual covariance matrix of asset returns.
$c \in \mathbb{R}_+^N$	Linear execution-cost rates.
$\Lambda \succeq 0$	Quadratic market-impact matrix.
$L \in \mathbb{R}^{S\times N}$	Scenario-loss matrix; row s contains the loss of every asset in scenario s.
τ_s, H_s	Soft warning threshold and hard loss limit for scenario s.
$h_s \geq 0$	Auxiliary variable measuring warning excess.

3 From dollar holdings to portfolio weights

Let total portfolio value be V dollars and let x_i be the dollar amount invested in asset i. The portfolio weight is

$$w_i = \frac{x_i}{V}. \tag{1}$$

Because all capital must be assigned,

$$\sum_{i=1}^{N} w_i = 1 \quad \Longleftrightarrow \quad \mathbf{1}^\mathsf{T} w = 1. \tag{2}$$

Long-only investing gives $w_i \geq 0$. Asset-specific caps u_i give

$$0 \leq w_i \leq u_i. \tag{3}$$

This weight representation makes portfolios of different dollar sizes directly comparable.

4 Exact trade accounting

A naive expression for turnover is $\sum_i |w_i - w_i^0|$. Absolute values are inconvenient inside a smooth numerical optimizer. The project introduces nonnegative purchases p_i and sales n_i:

$$w = w^0 + p - n, \qquad p \geq 0, \quad n \geq 0. \tag{4}$$

For each asset,

$$w_i - w_i^0 = p_i - n_i. \tag{5}$$

If transaction costs are positive, buying and selling the same asset simultaneously is dominated: reducing both p_i and n_i by the smaller amount leaves w_i unchanged and lowers cost. Therefore, at an optimum,

$$p_i + n_i = \left| w_i - w_i^0 \right|. \tag{6}$$

Gross turnover and one-way turnover are then

$$T_{\text{gross}} = \sum_{i=1}^{N} (p_i + n_i) = \left\| w - w^0 \right\|_1, \tag{7}$$

$$T_{\text{one-way}} = \frac{1}{2} T_{\text{gross}}. \tag{8}$$

The final policy uses a 50% gross-turnover cap:

$$\mathbf{1}^{\mathsf{T}}(p + n) \leq 0.50. \tag{9}$$

5 Expected return: growth plus income

Asset i has expected capital appreciation g_i and expected income yield y_i. Linearity of expectation gives

$$G(w) = \mathbb{E}[\text{growth}] = g^{\mathsf{T}} w, \tag{10}$$

$$Y(w) = \mathbb{E}[\text{income}] = y^{\mathsf{T}} w, \tag{11}$$

$$R(w) = G(w) + Y(w) = (g + y)^{\mathsf{T}} w. \tag{12}$$

This separation is useful because an investor can prefer income without pretending that dividends and price appreciation are the same economic goal.

6 Covariance risk derived from random returns

Let the random return vector be $r \in \mathbb{R}^N$ with mean $\mu = \mathbb{E}[r]$ and covariance

$$\Sigma = \mathbb{E}\left[(r - \mu)(r - \mu)^{\mathsf{T}} \right]. \tag{13}$$

Portfolio return is $r_p = w^\mathsf{T} r$. Its variance is

$$\begin{aligned}\mathrm{Var}(r_p) &= \mathbb{E}\left[(w^\mathsf{T} r - w^\mathsf{T}\mu)^2\right] &(14)\\ &= \mathbb{E}\left[w^\mathsf{T}(r-\mu)(r-\mu)^\mathsf{T} w\right] &(15)\\ &= w^\mathsf{T}\mathbb{E}\left[(r-\mu)(r-\mu)^\mathsf{T}\right] w &(16)\\ &= w^\mathsf{T}\Sigma w. &(17)\end{aligned}$$

Annual volatility is

$$\sigma(w) = \sqrt{w^\mathsf{T}\Sigma w}. \tag{18}$$

Because Σ is positive semidefinite, $w^\mathsf{T}\Sigma w \geq 0$, and the variance term is convex.

7 Concentration and effective holdings

The Herfindahl concentration measure is

$$C(w) = \sum_i w_i^2 = w^\mathsf{T} w. \tag{19}$$

Its reciprocal is the effective number of equally weighted holdings:

$$N_{\text{eff}}(w) = \frac{1}{\sum_i w_i^2}. \tag{20}$$

For example, ten equal 10% positions have $C = 10(0.1)^2 = 0.1$ and $N_{\text{eff}} = 10$. Concentration is not a complete diversification measure because ETFs can overlap, but it is an auditable first guardrail.

8 Implementation costs

8.1 Linear execution cost

If c_i is the proportional cost of trading asset i, the direct cost is

$$C_{\text{lin}}(p, n) = c^\mathsf{T}(p+n). \tag{21}$$

This includes commissions and spread-like costs and is linear in trade size.

8.2 Quadratic market impact

Large trades become progressively more expensive. Let $\Delta w = w - w^0$. The project models nonlinear impact by

$$C_{\text{imp}}(w) = \Delta w^\mathsf{T}\Lambda\Delta w, \tag{22}$$

where $\Lambda \succeq 0$. A diagonal Λ means each asset has its own impact coefficient; off-diagonal elements can represent interacting trades.

9 Scenario losses, warning thresholds, and hard limits

For scenario s, let $\ell_s \in \mathbb{R}^N$ contain asset losses. Portfolio loss is

$$L_s(w) = \ell_s^\mathsf{T} w. \tag{23}$$

A hard policy requires

$$L_s(w) \le H_s. \tag{24}$$

A softer warning threshold $\tau_s < H_s$ creates a graded penalty before the hard limit is reached. Define

$$h_s = \max\{L_s(w) - \tau_s, 0\}. \tag{25}$$

This can be represented with linear constraints

$$h_s \ge L_s(w) - \tau_s, \qquad h_s \ge 0, \tag{26}$$

and a convex quadratic penalty

$$C_{\text{scenario}}(w, h) = \sum_{s=1}^{S} \omega_s h_s^2. \tag{27}$$

The distinction is important: a portfolio can pass every hard limit while still breaching one or more warning levels.

10 Trade-capacity constraints

Let ADV_i be average daily dollar volume, D the number of execution days, ρ the allowed participation rate, and V portfolio value. The maximum executable dollar trade is $D\rho\,\mathrm{ADV}_i$. Dividing by V gives the weight capacity

$$p_i + n_i \le \frac{D\rho\,\mathrm{ADV}_i}{V}. \tag{28}$$

This prevents the optimizer from assuming that an arbitrarily large trade can be executed at the same modeled cost.

11 Class, factor, return, and income guardrails

Let $a_k \in \{0,1\}^N$ identify members of asset class k. Class exposure is $a_k^\mathsf{T} w$, constrained by

$$\underline{a}_k \le a_k^\mathsf{T} w \le \bar{a}_k. \tag{29}$$

If $B \in \mathbb{R}^{N\times F}$ is a factor-loading matrix, factor f exposure is $B_{:,f}^\mathsf{T} w$ and can be banded similarly. Minimum expected return and income are

$$(g + y)^\mathsf{T} w \ge R_{\min}, \qquad y^\mathsf{T} w \ge Y_{\min}. \tag{30}$$

12 The complete continuous optimization problem

The final minimization objective is

$$\min_{w,p,n,h} \quad F = -\alpha_g g^\top w - \alpha_y y^\top w + \lambda_v w^\top \Sigma w + \lambda_c w^\top w \tag{31}$$

$$+ \lambda_L c^\top (p+n) + \lambda_I (w - w^0)^\top \Lambda (w - w^0) + \lambda_s \sum_s \omega_s h_s^2. \tag{32}$$

It is subject to the budget, trade accounting, bounds, class/factor bands, return and income floors, turnover, liquidity capacity, warning-slack, and hard-scenario constraints derived above.

Why the model is solvable. Every hard constraint is linear. Every quadratic penalty uses a positive-semidefinite matrix or a squared nonnegative slack. Therefore the mathematical core is a convex quadratic program when the active support is fixed. This property is what later allows independent CVXPY and KKT validation.

13 Worked three-asset example

Suppose the current portfolio is $w^0 = (0.50, 0.30, 0.20)$ and the proposed portfolio is $w = (0.40, 0.35, 0.25)$. Then

$$\Delta w = (-0.10, 0.05, 0.05), \tag{33}$$

$$p = (0, 0.05, 0.05), \tag{34}$$

$$n = (0.10, 0, 0), \tag{35}$$

$$T_{\text{gross}} = 0.10 + 0.05 + 0.05 = 0.20. \tag{36}$$

If expected total returns are $(6\%, 4\%, 3\%)$, the portfolio expectation is

$$R(w) = 0.40(0.06) + 0.35(0.04) + 0.25(0.03) = 4.55\%. \tag{37}$$

If a stress vector is $(20\%, 5\%, 2\%)$, the stress loss is

$$L(w) = 0.40(0.20) + 0.35(0.05) + 0.25(0.02) = 10.25\%. \tag{38}$$

A warning at 9% gives $h = 1.25\%$; a hard limit at 12% is still satisfied. This miniature example shows how the same portfolio can be hard-compliant but warning-breaching.

14 How this formulation becomes the project architecture

1. Step 3 estimates g, y, Σ, c, Λ, liquidity, classes, and factor loadings.
2. Step 4 solves progressively richer versions of the exact continuous problem.
3. Step 5 maps user preferences to the coefficients α and λ.
4. Step 5Q selects a reduced active support with QAOA, then returns to this exact model.
5. Steps 6 and 7 audit and independently validate the resulting portfolios.
6. Step 10 ranks only zero-hard-breach candidates.

15 Reproduction checklist

A faithful implementation must verify: (i) weights sum to one; (ii) $w = w^0 + p - n$; (iii) turnover and capacity are calculated from $p + n$; (iv) Σ and Λ are positive semidefinite; (v) scenario slacks equal the exact positive part; (vi) every reported candidate has a full constraint audit; and (vii) no binary support is reported before continuous reoptimization.

16 What Step 1 does and does not establish

Step 1 establishes a coherent, auditable economic optimization model. It does not establish that the input expectations are true, that future returns will match the model, or that a quantum algorithm is superior. Those questions belong to data validation, solver validation, and empirical testing in later stages.

A Detailed derivation of the standard quadratic program

This appendix rewrites the economic model in the standard form used by a quadratic-program solver. The purpose is not to change the economics, but to show exactly how each financial statement becomes a coefficient, a row of a matrix, or a variable bound.

Stack the decision variables as

$$x = \begin{bmatrix} w \\ p \\ n \\ h \end{bmatrix} \in \mathbb{R}^{3N+S}. \tag{39}$$

The objective can be written as

$$\frac{1}{2} x^\top H x + f^\top x + c_0. \tag{40}$$

To construct H, first expand the market-impact term:

$$(w - w^0)^\top \Lambda (w - w^0) = w^\top \Lambda w - 2(w^0)^\top \Lambda w + (w^0)^\top \Lambda w^0. \tag{41}$$

The last term is constant and can be dropped without changing the minimizer. The quadratic block for w is therefore

$$H_{ww} = 2\left(\lambda_v \Sigma + \lambda_c I + \lambda_I \Lambda\right), \tag{42}$$

because the solver convention contains the factor $1/2$. The scenario-slack block is

$$H_{hh} = 2\lambda_s \operatorname{diag}(\omega_1, \ldots, \omega_S). \tag{43}$$

All other blocks are zero in the baseline model. The linear coefficient on w is

$$f_w = -\alpha_g g - \alpha_y y - 2\lambda_I \Lambda w^0, \tag{44}$$

and the coefficients on purchases and sales are

$$f_p = f_n = \lambda_L c. \tag{45}$$

The coefficient on h is zero because its penalty is already quadratic. The omitted constant is

$$c_0 = \lambda_I (w^0)^\top \Lambda w^0. \tag{46}$$

It can be retained for objective reconciliation, but not for optimization.

The budget row is

$$\begin{bmatrix} \mathbf{1}^\top & 0 & 0 & 0 \end{bmatrix} x = 1. \tag{47}$$

The trade-accounting equations are, asset by asset,

$$w_i - p_i + n_i = w_i^0. \tag{48}$$

Together these produce $N + 1$ equality rows. Asset bounds, turnover, class bands, factor bands, return and income floors, trade capacities, warning slacks, and hard scenario limits become inequality rows. For example, the warning-slack definition

$$h_s \geq \ell_s^\top w - \tau_s \tag{49}$$

is entered as

$$\ell_s^\top w - h_s \leq \tau_s. \tag{50}$$

This explicit matrix construction is what Step 7 independently rebuilds.

B Why purchases and sales reproduce the absolute trade exactly

The model does not explicitly impose $p_i n_i = 0$. Nevertheless, simultaneous positive buying and selling is suboptimal when transaction cost is increasing in $p_i + n_i$. Suppose $p_i > 0$ and $n_i > 0$. Let $\delta = \min(p_i, n_i) > 0$ and define

$$p_i' = p_i - \delta, \qquad n_i' = n_i - \delta. \tag{51}$$

Then

$$p_i' - n_i' = p_i - n_i, \tag{52}$$

so the final weight is unchanged, but

$$p_i' + n_i' = p_i + n_i - 2\delta < p_i + n_i. \tag{53}$$

Linear cost and any nondecreasing turnover penalty fall. Therefore a cost-minimizing optimum cannot have simultaneous purchases and sales in the same asset, except for numerical noise. This proves

$$p_i + n_i = |w_i - w_i^0| \tag{54}$$

at the optimum.

C Why the scenario slack equals the positive part

The constraints require

$$h_s \geq L_s(w) - \tau_s, \qquad h_s \geq 0. \tag{55}$$

The objective has a strictly nondecreasing penalty in $h_s \geq 0$. If h_s were larger than both lower bounds, reducing it would preserve feasibility and lower the objective. Thus the optimum chooses the smallest feasible value:

$$h_s = \max\{L_s(w) - \tau_s, 0\}. \tag{56}$$

This is an exact epigraph representation, not an approximation.

D Convexity proof term by term

A function $x \mapsto x^\mathsf{T} M x$ is convex when $M \succeq 0$. The covariance matrix is positive semidefinite by definition. The identity matrix is positive definite. The impact matrix is constructed positive semidefinite. The scenario penalty is a weighted sum of squares with nonnegative weights. Linear rewards and costs do not affect convexity. The feasible region is an intersection of affine equalities, linear inequalities, and boxes, all convex sets. Hence the fixed-support continuous problem is convex.

Convexity has a practical consequence. If a feasible solution satisfies the Karush–Kuhn–Tucker conditions to numerical tolerance, it is a global optimum of the declared model. It does not prove that the expected returns are correct, but it does prove that the solver did not merely stop at a bad local minimum.

E Coefficient scaling and dimensional consistency

Return, variance, cost, concentration, and scenario penalties have different natural magnitudes. Directly adding them would allow the largest numerical scale to dominate unintentionally. The project therefore calibrates characteristic scales $s_j > 0$ and uses dimensionless quantities

$$\widetilde{m}_j(w) = \frac{m_j(w)}{s_j}. \tag{57}$$

A preference coefficient then expresses importance rather than unit conversion. A reproducible implementation must save the scales, because changing a scale changes the effective economic trade-off even when the visible preference score is unchanged.

A useful audit is to print each weighted objective contribution at the solution. If one term is orders of magnitude larger than all others, either that dominance is intentional or the scaling must be reviewed.

F Worked covariance example

Consider three assets with annual volatilities 10%, 15%, and 20%, and correlations

$$\rho = \begin{bmatrix} 1 & 0.2 & -0.1 \\ 0.2 & 1 & 0.4 \\ -0.1 & 0.4 & 1 \end{bmatrix}. \tag{58}$$

Let $D = \operatorname{diag}(0.10, 0.15, 0.20)$. The covariance matrix is

$$\Sigma = D\rho D. \tag{59}$$

For weights $w = (0.40, 0.35, 0.25)$, variance is

$$\sigma_p^2 = w^\mathsf{T} \Sigma w, \tag{60}$$

and volatility is $\sqrt{\sigma_p^2}$. The off-diagonal covariance terms matter because they can either amplify or offset stand-alone risk. This is why choosing the individually least volatile assets is not generally the same as choosing the least volatile portfolio.

G Constraint taxonomy for a reviewer

Constraint family	Financial purpose	Mathematical form
Budget	Invest all capital	$\mathbf{1}^\top w = 1$
Long-only and caps	Prevent shorting and concentration	$0 \le w_i \le u_i$
Trade accounting	Link target to incumbent	$w = w^0 + p - n$
Turnover	Limit portfolio disruption	$\mathbf{1}^\top (p + n) \le T$
Liquidity capacity	Keep trades executable	$p_i + n_i \le q_i$
Class bands	Preserve strategic diversification	$\underline{a}_k \le a_k^\top w \le \overline{a}_k$
Factor bands	Limit hidden systematic bets	$\underline{b}_f \le B_{:,f}^\top w \le \overline{b}_f$
Return and income floors	Preserve mandate economics	$(g + y)^\top w \ge R_{\min}$, $y^\top w \ge Y_{\min}$
Scenario hard limits	Prevent unacceptable modeled losses	$\ell_s^\top w \le H_s$
Warning slacks	Penalize risk before hard failure	$h_s \ge \ell_s^\top w - \tau_s$, $h_s \ge 0$

H Implementation-to-equation map

A reproducing researcher should be able to trace every equation to code and every reported metric back to primitive data. The recommended sequence is:

1. Build a typed data object containing tickers, expected growth, income, covariance, class labels, factor loadings, costs, and liquidity.
2. Validate dimensions, missing values, symmetry, and positive-semidefinite matrices.
3. Build the variable vector (w, p, n, h) and the objective blocks.
4. Add constraints one family at a time and unit-test each family on a deliberately violating portfolio.
5. Solve the convex problem and recompute every metric from the returned weights, rather than trusting solver auxiliary values.
6. Store the complete constraint table with value, lower bound, upper bound, slack, and pass/fail status.
7. Export the coefficients and configuration so Step 7 can rebuild the model independently.

I Minimum mathematical tests

- **Budget identity:** random feasible portfolios must sum to one within tolerance.
- **Trade identity:** verify $w - w^0 = p - n$ and $p, n \ge 0$.
- **Turnover identity:** verify $\sum (p + n) = \sum |w - w^0|$ at the solution.
- **Variance agreement:** compare $w^\top \Sigma w$ with volatility squared.
- **PSD check:** the minimum eigenvalue of Σ and Λ must be above a small negative tolerance.
- **Scenario agreement:** compare stored slacks with $\max(Lw - \tau, 0)$.

- **Objective reconciliation:** sum every reward and penalty independently and match the reported objective.
- **Constraint audit:** no candidate may be called feasible unless every applicable hard row passes.